

Machine behaviour

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Machines powered by artificial intelligence increasingly mediate our social, cultural, economic and political interactions. Understanding the behaviour of artificial intelligence systems is essential to our ability to control their actions, reap their benefits and minimize their harms. Here we argue that this necessitates a broad scientific research agenda to study machine behaviour that incorporates and expands upon the discipline of computer science and includes insights from across the sciences. We first outline a set of questions that are fundamental to this emerging field and then explore the technical, legal and institutional constraints on the study of machine behaviour.

In his landmark 1969 book *Sciences of the Artificial*¹, Nobel Laureate Herbert Simon wrote: “Natural science is knowledge about natural objects and phenomena. We ask whether there cannot also be ‘artificial’ science—knowledge about artificial objects and phenomena.” In line with Simon’s vision, we describe the emergence of an interdisciplinary field of scientific study. This field is concerned with the scientific study of intelligent machines, not as engineering artefacts, but as a class of actors with particular behavioural patterns and ecology. This field overlaps with, but is distinct from, computer science and robotics. It treats machine behaviour empirically. This is akin to how ethology and behavioural ecology study animal behaviour by integrating physiology and biochemistry—intrinsic properties—with the study of ecology and evolution—properties shaped by the environment. Animal and human behaviours cannot be fully understood without the study of the contexts in which behaviours occur. Machine behaviour similarly cannot be fully understood without the integrated study of algorithms and the social environments in which algorithms operate².

At present, the scientists who study the behaviours of these virtual and embodied artificial intelligence (AI) agents are predominantly the same scientists who have created the agents themselves (throughout we use the term ‘AI agents’ liberally to refer to both complex and simple algorithms used to make decisions). As these scientists create agents to solve particular tasks, they often focus on ensuring the agents fulfil their intended function (although these respective fields are much broader than the specific examples listed here). For example, AI agents should meet a benchmark of accuracy in document classification, facial recognition or visual object detection. Autonomous cars must navigate successfully in a variety of weather conditions; game-playing agents must defeat a variety of human or machine opponents; and data-mining agents must learn



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which individuals to target in advertising campaigns on social media.

These AI agents have the potential to augment human welfare and well-being in many ways. Indeed, that is typically the vision of their creators. But a

broader consideration of the behaviour of AI agents is now critical. AI agents will increasingly integrate into our society and are already involved in a variety of activities, such as credit scoring, algorithmic trading, local policing, parole decisions, driving, online dating and drone warfare^{3,4}. Commentators and scholars from diverse fields—including, but not limited to, cognitive systems engineering, human computer interaction, human factors, science, technology and society, and safety engineering—are raising the alarm about the broad, unintended consequences of AI agents that can exhibit behaviours and produce downstream societal effects—both positive and negative—that are unanticipated by their creators^{5–8}.

In addition to this lack of predictability surrounding the consequences of AI, there is a fear of the potential loss of human oversight over intelligent machines⁵ and of the potential harms that are associated with the increasing use of machines for tasks that were once performed directly by humans⁹. At the same time, researchers describe the benefits that AI agents can offer society by supporting and augmenting human decision-making^{10,11}. Although discussions of these issues have led to many important insights in many separate fields of academic inquiry¹², with some highlighting safety challenges of autonomous systems¹³ and others studying the implications in fairness, accountability and transparency (for example, the ACM conference on fairness, accountability and transparency (<https://fatconference.org/>)), many questions remain.

This Review frames and surveys the emerging interdisciplinary field of machine behaviour: the scientific study of behaviour exhibited by

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intelligent machines. Here we outline the key research themes, questions and landmark research studies that exemplify this field. We start by providing background on the study of machine behaviour and the necessarily interdisciplinary nature of this science. We then provide a framework for the conceptualization of studies of machine behaviour. We close with a call for the scientific study of machine and human–machine ecologies and discuss some of the technical, legal and institutional barriers that are faced by researchers in this field.

Motivation for the study of machine behaviour

There are three primary motivations for the scientific discipline of machine behaviour. First, various kinds of algorithms operate in our society, and algorithms have an ever-increasing role in our daily activities. Second, because of the complex properties of these algorithms and the environments in which they operate, some of their attributes and behaviours can be difficult or impossible to formalize analytically. Third, because of their ubiquity and complexity, predicting the effects of intelligent algorithms on humanity—whether positive or negative—poses a substantial challenge.

Ubiquity of algorithms

The current prevalence of diverse algorithms in society is unprecedented⁵ (Fig. 1). News-ranking algorithms and social media bots influence the information seen by citizens^{14–18}. Credit-scoring algorithms determine loan decisions^{19–22}. Online pricing algorithms shape the cost of products differentially across consumers^{23–25}. Algorithmic trading software makes transactions in financial markets at rapid speed^{26–29}. Algorithms shape the dispatch and spatial patterns of local policing³⁰ and programs for algorithmic sentencing affect time served in the penal system⁷. Autonomous cars traverse our cities³¹, and ride-sharing algorithms alter the travel patterns of conventional vehicles³². Machines map our homes, respond to verbal commands³³ and perform regular household tasks³⁴. Algorithms shape romantic matches for online dating services^{35,36}. Machines are likely to increasingly substitute for humans in the raising of our young³⁷ and the care for our old³⁸. Autonomous agents are increasingly likely to affect collective behaviours, from group-wide coordination to sharing³⁹. Furthermore, although the prospect of developing autonomous weapons is highly controversial, with many in the field voicing their opposition^{6,40}, if such weapons end up being deployed, then machines could determine who lives and who dies in armed conflicts^{41,42}.

Complexity and opacity of algorithms

The extreme diversity of these AI systems, coupled with their ubiquity, would by itself ensure that studying the behaviour of such systems poses a formidable challenge, even if the individual algorithms themselves were relatively simple. The complexity of individual AI agents is currently high and rapidly increasing. Although the code for specifying the architecture and training of a model can be simple, the results can be very complex, oftentimes effectively resulting in ‘black boxes’⁴³. They are given input and produce output, but the exact functional processes that generate these outputs are hard to interpret even to the very scientists who generate the algorithms themselves⁴⁴, although some progress in interpretability is being made^{45,46}. Furthermore, when systems learn from data, their failures are linked to imperfections in the data or how data was collected, which has led some to argue for adapted reporting mechanisms for datasets⁴⁷ and models⁴⁸. The dimensionality and size of data add another layer of complexity to understanding machine behaviour⁴⁹.

Further complicating this challenge is the fact that much of the source code and model structure for the most frequently used algorithms in society is proprietary, as are the data on which these systems are trained. Industrial secrecy and legal protection of intellectual property often surround source code and model structure. In many settings, the only factors that are publicly observable about industrial AI systems are their inputs and outputs.

Even when available, the source code or model structure of an AI agent can provide insufficient predictive power over its output. AI

agents can also demonstrate novel behaviours through their interaction with the world and other agents that are impossible to predict with precision⁵⁰. Even when the analytical solutions are mathematically describable, they can be so lengthy and complex as to be indecipherable^{51,52}. Furthermore, when the environment is changing—perhaps as a result of the algorithm itself—anticipating and analysing behaviour is made much harder.

Algorithms’ beneficial and detrimental effect on humanity

The ubiquity of algorithms, coupled with their increasing complexity, tends to amplify the difficulty of estimating the effects of algorithms on individuals and society. AI agents can shape human behaviours and societal outcomes in both intended and unintended ways. For example, some AI agents are designed to aid learning outcomes for children⁵³ and others are designed to assist older people^{38,54}. These AI systems may benefit their intended humans by nudging those humans into better learning or safer mobility behaviours. However, with the power to nudge human behaviours in positive or intended ways comes the risk that human behaviours may be nudged in costly or unintended ways—children could be influenced to buy certain branded products and elders could be nudged to watch certain television programs.

The way that such algorithmic influences on individual humans scale into society-wide effects, both positive and negative, is of critical concern. As an example, the exposure of a small number of individuals to political misinformation may have little effect on society as a whole. However, the effect of the insertion and propagation of such misinformation on social media may have more substantial societal consequences^{55–57}. Furthermore, issues of algorithmic fairness or bias^{58,59} have been already documented in diverse contexts, including computer vision⁶⁰, word embeddings^{61,62}, advertising⁶³, policing⁶⁴, criminal justice^{7,65} and social services⁶⁶. To address these issues, practitioners will sometimes be forced to make value trade-offs between competing and incompatible notions of bias^{58,59} or between human versus machine biases. Additional questions regarding the effect of algorithms remain, such as how online dating algorithms alter the societal institution of marriage^{35,36} and whether there are systemic effects of increasing interaction with intelligent algorithms on the stages and speed of human development⁵³. These questions become more complex in ‘hybrid systems’ composed of many machines and humans interacting and manifesting collective behaviour^{39,67}. For society to have input into and oversight of the downstream consequences of AI, scholars of machine behaviour must provide insights into how these systems work and the benefits, costs and trade-offs presented by the ubiquitous use of AI in society.

The interdisciplinary study of machine behaviour

To study machine behaviour—especially the behaviours of black box algorithms in real-world settings—we must integrate knowledge from across a variety of scientific disciplines (Fig. 2). This integration is currently in its nascent stages and has happened largely in an ad hoc fashion in response to the growing need to understand machine behaviour. Currently, the scientists who most commonly study the behaviour of machines are the computer scientists, roboticists and engineers who have created the machines in the first place. These scientists may be expert mathematicians and engineers; however, they are typically not trained behaviourists. They rarely receive formal instruction on experimental methodology, population-based statistics and sampling paradigms, or observational causal inference, let alone neuroscience, collective behaviour or social theory. Conversely, although behavioural scientists are more likely to possess training in these scientific methods, they are less likely to possess the expertise required to proficiently evaluate the underlying quality and appropriateness of AI techniques for a given problem domain or to mathematically describe the properties of particular algorithms.

Integrating scientific practices from across multiple fields is not easy. Up to this point, the main focus of those who create AI systems has been on crafting, implementing and optimizing intelligent systems to

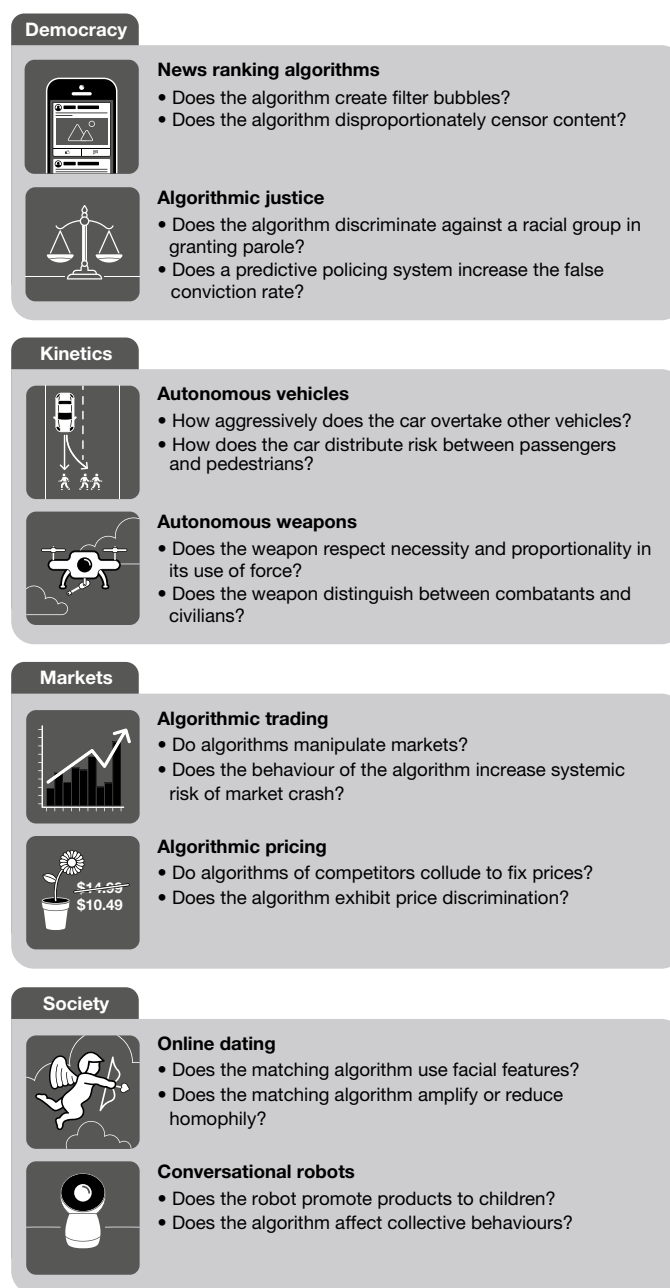


Fig. 1 | Examples of questions that fall into the domain of machine behaviour. Questions of concern to machine behaviour span a wide variety of traditional scientific disciplines and topics.

perform specialized tasks. Excellent progress has been made on benchmark tasks—including board games such as chess⁶⁸, checkers⁶⁹ and Go^{70,71}, card games such as poker⁷², computer games such as those on the Atari platform⁷³, artificial markets⁷⁴ and Robocup Soccer⁷⁵—as well as standardized evaluation data, such as the ImageNet data for object recognition⁷⁶ and the Microsoft Common Objects in Context data for image-captioning tasks⁷⁷. Success has also been achieved in speech recognition, language translation and autonomous locomotion. These benchmarks are coupled with metrics to quantify performance on standardized tasks^{78–81} and are used to improve performance, a proxy that enables AI builders to aim for better, faster and more-robust algorithms.

But methodologies aimed at maximized algorithmic performance are not optimal for conducting scientific observation of the properties and behaviours of AI agents. Rather than using metrics in the service of optimization against benchmarks, scholars of machine behaviour are interested in a broader set of indicators, much as social scientists explore a wide range of human behaviours in the realm of social, political or

economic interactions⁸². As such, scholars of machine behaviour spend considerable effort in defining measures of micro and macro outcomes to answer broad questions such as how these algorithms behave in different environments and whether human interactions with algorithms alter societal outcomes. Randomized experiments, observational inference and population-based descriptive statistics—methods that are often used in quantitative behavioural sciences—must be central to the study of machine behaviour. Incorporating scholars from outside of the disciplines that traditionally produce intelligent machines can provide knowledge of important methodological tools, scientific approaches, alternative conceptual frameworks and perspectives on the economic, social and political phenomena that machines will increasingly influence.

Type of question and object of study

Nikolaas Tinbergen, who won the 1973 Nobel Prize in Physiology or Medicine alongside Karl von Frisch and Konrad Lorenz for founding the field of ethology, identified four complementary dimensions of

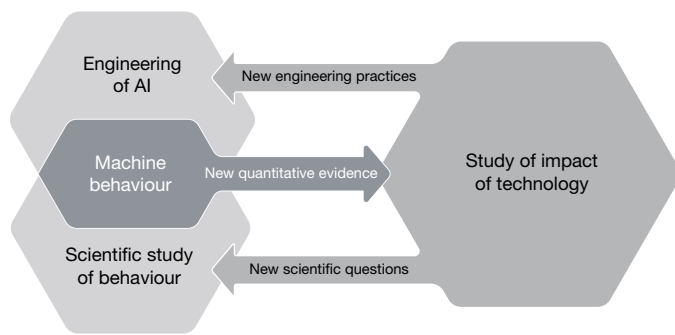


Fig. 2 | The interdisciplinarity of machine behaviour. Machine behaviour lies at the intersection of the fields that design and engineer AI systems and the fields that traditionally use scientific methods to study the behaviour of biological agents. The insights from machine behavioural studies provide quantitative evidence that can help to inform those fields that study the potential effects of technology on social and technological systems. In turn, those fields can provide useful engineering practices and scientific questions to fields that examine machine behaviours. Finally, the scientific study of behaviour helps AI scholars to make more precise statements about what AI systems can and cannot do.

analysis that help to explain animal behaviour⁸³. These dimensions concern questions of the function, mechanism, development and evolutionary history of a behaviour and provide an organizing framework for the study of animal and human behaviour. For example, this conceptualization distinguishes the study of how a young animal or human develops a type of behaviour from the evolutionary trajectory that selected for such behaviour in the population. The goal of these distinctions is not division but rather integration. Although it is not wrong to say that, for example, a bird's song is explained by learning or by its specific evolutionary history, a complete understanding of the song will require both.

Despite fundamental differences between machines and animals, the behavioural study of machines can benefit from a similar classification. Machines have mechanisms that produce behaviour, undergo development that integrates environmental information into behaviour, produce functional consequences that cause specific machines to become more or less common in specific environments and embody evolutionary histories through which past environments and human decisions continue to influence machine behaviour. Scholars of computer science have already achieved substantial gains in understanding the mechanisms and development of AI systems, although many questions remain. Relatively less emphasis has been placed on the function and evolution of AI systems. We discuss these four topics in the next subsections and provide Fig. 3 as a summary⁸⁴.

Mechanisms for generating behaviour

The proximate causes of a machine's behaviour have to do with how the behaviour is observationally triggered and generated in specific environments. For example, early algorithmic trading programs used simple rules to trigger buying and selling behaviour⁸⁵. More sophisticated agents may compute strategies based on adaptive heuristics or explicit maximization of expected utility⁸⁶. The behaviour of a reinforcement learning algorithm that plays poker could be attributed to the particular way in which it represents the state space or evaluates the game tree⁷², and so on.

A mechanism depends on both an algorithm and its environment. A more sophisticated agent, such as a driverless car, may exhibit particular driving behaviour—for example, lane switching, overtaking or signalling to pedestrians. These behaviours would be generated according to the algorithms that construct driving policies⁸⁷ and are also shaped fundamentally by features of the perception and actuation system of the car, including the resolution and accuracy of its object detection and classification system, and the responsiveness and accuracy of its steering, among other factors. Because many current AI systems are

derived from machine learning methods that are applied to increasingly complex data, the study of the mechanism behind a machine's behaviour, such as those mentioned above, will require continued work on interpretability methods for machine learning^{46,88,89}.

Development of behaviour

In the study of animal or human behaviour, development refers to how an individual acquires a particular behaviour—for example, through imitation or environmental conditioning. This is distinct from longer-term evolutionary changes.

In the context of machines, we can ask how machines acquire (develop) a specific individual or collective behaviour. Behavioural development could be directly attributable to human engineering or design choices. Architectural design choices made by the programmer (for example, the value of a learning rate parameter, the acquisition of the representation of knowledge and state, or a particular wiring of a convolutional neural network) determine or influence the kinds of behaviours that the algorithm exhibits. In a more complex AI system, such as a driverless car, the behaviour of the car develops over time, from software development and changing hardware components that engineers incorporate into its overall architecture. Behaviours can also change as a result of algorithmic upgrades pushed to the machine by its designers after deployment.

A human engineer may also shape the behaviour of the machine by exposing it to particular training stimuli. For instance, many image and text classification algorithms are trained to optimize accuracy on a specific set of datasets that were manually labelled by humans. The choice of dataset—and those features it represents^{60,61}—can substantially influence the behaviour exhibited by the algorithm.

Finally, a machine may acquire behaviours through its own experience. For instance, a reinforcement learning agent trained to maximize long-term profit can learn peculiar short-term trading strategies based on its own past actions and concomitant feedback from the market⁹⁰. Similarly, product recommendation algorithms make recommendations based on an endless stream of choices made by customers and update their recommendations accordingly.

Function

In the study of animal behaviour, adaptive value describes how a behaviour contributes to the lifetime reproductive fitness of an animal. For example, a particular hunting behaviour may be more or less successful than another at prolonging the animal's life and, relatedly, the number of mating opportunities, resulting offspring born and the probable reproductive success of the offspring. The focus on function helps us to understand why some behavioural mechanisms spread and persist while others decline and vanish. Function depends critically on the fit of the behaviour to environment.

In the case of machines, we may talk of how the behaviour fulfils a contemporaneous function for particular human stakeholders. The human environment creates selective forces that may make some machines more common. Behaviours that are successful ('fitness' enhancing) get copied by developers of other software and hardware or are sometimes engineered to propagate among the machines themselves. These dynamics are ultimately driven by the success of institutions—such as corporations, hospitals, municipal governments and universities—that build or use AI. The most obvious example is provided by algorithmic trading, in which successful automated trading strategies could be copied as their developers move from company to company, or are simply observed and reverse-engineered by rivals.

These forces can produce unanticipated effects. For example, objectives such as maximizing engagement on a social media site may lead to so-called filter bubbles⁹¹, which may increase political polarization or, without careful moderation, could facilitate the spread of fake news. However, websites that do not optimize for user engagement may not be as successful in comparison with ones that do, or may go out of business altogether. Similarly, in the absence of external regulation, autonomous cars that do not prioritize the safety of their own passengers may be

Type of question	Object of study	
	Dynamic view Explanation of current form in terms of a historical sequence	Static view Explanation of the current behaviour of a machine
Proximate view How a particular type of machine functions	Development (ontogeny) Developmental explanations of how a type of machine acquires its behaviour, from deliberate engineering and supervised learning based on specific benchmarks, to online learning and reinforcement learning in a particular environment.	Mechanism (causation) Mechanistic explanations for what the behaviour is, and how it is constructed, including computational mechanisms or external stimuli that trigger it.
Ultimate (evolutionary) view Why a type of machine evolved the behaviours it has	Evolution (phylogeny) Incentives and market forces that describe why the behaviour evolved and spread, whether by programming or learning, subject to computational and institutional constraints.	Function (adaptive value) The consequences of the machine's behaviour in the current environment that cause it to persist, either by appeal for particular stakeholders (such as users or companies) or fit to some other aspect of the environment.

Fig. 3 | Tinbergen's type of question and object of study modified for the study of machine behaviour. The four categories Tinbergen proposed for the study of animal behaviour can be adapted to the study of machine behaviour^{83,84}. Tinbergen's framework proposes two types of question,

less attractive to consumers, leading to fewer sales³¹. Sometimes the function of machine behaviour is to cope with the behaviour of other machines. Adversarial attacks—synthetic inputs that fool a system into producing an undesired output^{44,92–94}—on AI systems and the subsequent responses of those who develop AI to these attacks⁹⁵ may produce complex predator–prey dynamics that are not easily understood by studying each machine in isolation.

These examples highlight how incentives created by external institutions and economic forces can have indirect but substantial effects on the behaviours exhibited by machines⁹⁶. Understanding the interaction between these incentives and AI is relevant to the study of machine behaviour. These market dynamics would, in turn, interact with other processes to produce evolution among machines and algorithms.

Evolution

In the study of animal behaviour, phylogeny describes how a behaviour evolved. In addition to its current function, behaviour is influenced by past selective pressures and previously evolved mechanisms. For example, the human hand evolved from the fin of a bony fish. Its current function is no longer for swimming, but its internal structure is explained by its evolutionary history. Non-selective forces, such as migration and drift, also have strong roles in explaining relationships among different forms of behaviour.

In the case of machines, evolutionary history can also generate path dependence, explaining otherwise puzzling behaviour. At each step, aspects of the algorithms are reused in new contexts, both constraining future behaviour and making possible additional innovations. For example, early choices about microprocessor design continue to influence modern computing, and traditions in algorithm design—such as neural networks and Bayesian state–space models—build in many assumptions and guide future innovations by making some new algorithms easier to access than others. As a result, some algorithms may attend to certain features and ignore others because those features were important in early successful applications. Some machine behaviour may spread because it is 'evolvable'—easy to modify and robust to perturbations—similar to how some traits of animals may be common because they facilitate diversity and stability⁹⁷.

Machine behaviour evolves differently from animal behaviour. Most animal inheritance is simple—two parents, one transmission event. Algorithms are much more flexible and they have a designer with an objective in the background. The human environment strongly

influences how algorithms evolve by changing their inheritance system. AI replication behaviour may be facilitated through a culture of open source sharing of software, the details of network architecture or underlying training datasets. For instance, companies that develop software for driverless cars may share enhanced open source libraries for object detection or path planning as well as the training data that underlie these algorithms to enable safety-enhancing software to spread throughout the industry. It is possible for a single adaptive 'mutation' in the behaviour of a particular driverless car to propagate instantly to millions of other cars through a software update. However, other institutions apply limits as well. For example, software patents may impose constraints on the copying of particular behavioural traits. And regulatory constraints—such as privacy protection laws—can prevent machines from accessing, retaining or otherwise using particular information in their decision-making. These peculiarities highlight the fact that machines may exhibit very different evolutionary trajectories, as they are not bound by the mechanisms of organic evolution.

Scale of inquiry

With the framework outlined above and in Fig. 3, we now catalogue examples of machine behaviour at the three scales of inquiry: individual machines, collectives of machines and groups of machines embedded in a social environment with groups of humans in hybrid or heterogeneous systems³⁹ (Fig. 4). Individual machine behaviour emphasizes the study of the algorithm itself, collective machine behaviour emphasizes the study of interactions between machines and hybrid human–machine behaviour emphasizes the study of interactions between machines and humans. Here we can draw an analogy to the study of a particular species, the study of interactions among members of a species and the interactions of the species with their broader environment. Analyses at any of these scales may address any or all of the questions described in Fig. 3.

Individual machine behaviour

The study of the behaviour of individual machines focuses on specific intelligent machines by themselves. Often these studies focus on properties that are intrinsic to the individual machines and that are driven by their source code or design. The fields of machine learning and software engineering currently conduct the majority of these studies. There are two general approaches to the study of individual machine behaviour. The first focuses on profiling the set of behaviours of any

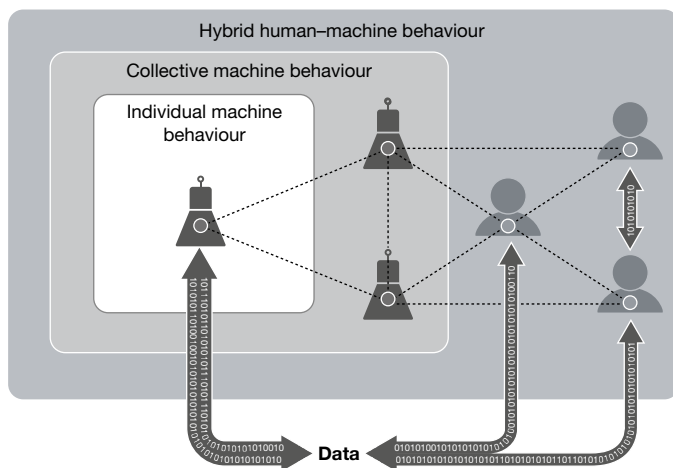


Fig. 4 | Scale of inquiry in the machine behaviour ecosystem. AI systems represent the amalgamation of humans, data and algorithms. Each of these domains influences the other in both well-understood and unknown ways. Data—filtered through algorithms created by humans—influences individual and collective machine behaviour. AI systems are trained on the data, in turn influencing how humans generate data. AI systems collectively interact with and influence one another. Human interactions can be altered by the introduction of these AI systems. Studies of machine behaviour tend to occur at the individual, the collective or the hybrid human-machine scale of inquiry.

specific machine agent using a within-machine approach, comparing the behaviour of a particular machine across different conditions. The second, a between-machine approach, examines how a variety of individual machine agents behave in the same condition.

A within-machine approach to the study of individual machine behaviours investigates questions such as whether there are constants that characterize the within-machine behaviour of any particular AI across a variety of contexts, how the behaviour of a particular AI progresses over time in the same, or different, environments and which environmental factors lead to the expression of particular behaviours by machines.

For instance, an algorithm may only exhibit certain behaviours if trained on particular underlying data^{98–100} (Fig. 3). Then, the question becomes whether or not an algorithm that scores probability of recidivism in parole decisions⁷ would behave in unexpected ways when presented with evaluation data that diverge substantially from its training data. Other studies related to the characterization of within-machine behaviour include the study of individual robotic recovery behaviours^{101,102}, the ‘cognitive’ attributes of algorithms and the utility of using techniques from psychology in the study of algorithmic behaviour¹⁰³, and the examination of bot-specific characteristics such as those designed to influence human users¹⁰⁴.

The second approach to the study of individual machine behaviour examines the same behaviours as they vary between machines. For example, those interested in examining advertising behaviours of intelligent agents^{63,105,106} may investigate a variety of advertising platforms (and their underlying algorithms) and examine the between-machine effect of performing experiments with the same set of advertising inputs across platforms. The same approach could be used for investigations of dynamic pricing algorithms^{23,24,32} across platforms. Other between-machine studies might look at the different behaviours used by autonomous vehicles in their overtaking patterns or at the varied foraging behaviours exhibited by search and rescue drones¹⁰⁷.

Collective machine behaviour

In contrast the study of the behaviour of individual machines, the study of collective machine behaviour focuses on the interactive and system-wide behaviours of collections of machine agents. In some cases, the implications of individual machine behaviour may make little sense

until the collective level is considered. Some investigations of these systems have been inspired by natural collectives, such as swarms of insects, or mobile groups, such as flocking birds or schooling fish. For example, animal groups are known to exhibit both emergent sensing of complex environmental features¹⁰⁸ and effective consensus decision-making¹⁰⁹. In both scenarios, groups exhibit an awareness of the environment that does not exist at the individual level. Fields such as multi-agent systems and computational game theory provide useful examples of the study of this area of machine behaviour.

Robots that use simple algorithms for local interactions between bots can nevertheless produce interesting behaviour once aggregated into large collectives. For example, scholars have examined the swarm-like properties of microrobots that combine into aggregations that resemble swarms found in systems of biological agents^{110,111}. Additional examples include the collective behaviours of algorithms both in the laboratory (in the Game of Life¹¹²) as well as in the wild (as seen in Wikipedia-editing bots¹¹³). Other examples include the emergence of novel algorithmic languages¹¹⁴ between communicating intelligent machines as well as the dynamic properties of fully autonomous transportation systems. Ultimately, many interesting questions in this domain remain to be examined.

The vast majority of work on collective animal behaviour and collective robotics has focused on how interactions among simple agents can create higher-order structures and properties. Although important, this neglects that fact that many organisms, and increasingly also AI agents⁷⁵, are sophisticated entities with behaviours and interactions that may not be well-characterized by simplistic representations. Revealing what extra properties emerge when interacting entities are capable of sophisticated cognition remains a key challenge in the biological sciences and may have direct parallels in the study of machine behaviour. For example, similar to animals, machines may exhibit ‘social learning’. Such social learning does not need be limited to machines learning from machines, but we may expect machines to learn from humans, and vice versa for humans to learn from the behaviour of machines. The feedback processes introduced may fundamentally alter the accumulation of knowledge, including across generations, directly affecting human and machine ‘culture’.

In addition, human-made AI systems do not necessarily face the same constraints as do organisms, and collective assemblages of machines provide new capabilities, such as instant global communication, that can lead to entirely new collective behavioural patterns. Studies in collective machine behaviour examine the properties of assemblages of machines as well as the unexpected properties that can emerge from these complex systems of interactions.

For example, some of the most interesting collective behaviour of algorithms has been observed in financial trading environments. These environments operate on tiny time scales, such that algorithmic traders can respond to events and each other ahead of any human trader¹¹⁵. Under certain conditions, high-frequency capabilities can produce inefficiencies in financial markets^{26,115}. In addition to the unprecedented response speed, the extensive use of machine learning, autonomous operation and ability to deploy at scale are all reasons to believe that the collective behaviour of machine trading may be qualitatively different than that of human traders. Furthermore, these financial algorithms and trading systems are necessarily trained on certain historic datasets and react to a limited variety of foreseen scenarios, leading to the question of how they will react to situations that are new and unforeseen in their design. Flash crashes are examples of clearly unintended consequences of (interacting) algorithms^{116,117}, leading to the question of whether algorithms could interact to create a larger market crisis.

Hybrid human-machine behaviour

Humans increasingly interact with machines¹⁶. They mediate our social interactions³⁹, shape the news^{14,17,55,56} and online information^{15,118} that we see, and form relationships with us that can alter our social systems. Because of their complexity, these hybrid human-machine systems

pose one of the most technically difficult yet simultaneously most important areas of study for machine behaviour.

Machines shape human behaviour

One of the most obvious—but nonetheless vital—domains of the study of machine behaviour concerns the ways in which the introduction of intelligent machines into social systems can alter human beliefs and behaviours. As in the introduction of automation to industrial processes¹¹⁹, intelligent machines can create social problems in the process of improving existing problems. Numerous problems and questions arise during this process, such as whether the matching algorithms that are used for online dating alter the distributional outcomes of the dating process or whether news-filtering algorithms alter the distribution of public opinion. It is important to investigate whether small errors in algorithms or the data that they use could compound to produce society-wide effects and how intelligent robots in our schools, hospitals¹²⁰ and care centres might alter human development¹²¹ and quality of life⁵⁴ and potentially affect outcomes for people with disabilities¹²².

Other questions in this domain relate to the potential for machines to alter the social fabric in more fundamental ways. For example, questions include to what extent and what ways are governments using machine intelligence to alter the nature of democracy, political accountability and transparency, or civic participation. Other questions include to what degree intelligent machines influence policing, surveillance and warfare, as well as how large of an effect bots have had on the outcomes of elections⁵⁶ and whether AI systems that aid in the formation of human social relationships can enable collective action.

Notably, studies in this area also examine how humans perceive the use of machines as decision aids^{8,123}, human preferences for and against making use of algorithms¹²⁴, and the degree to which human-like machines produce or reduce discomfort in humans^{39,125}. An important question in this area includes how humans respond to the increasing coproduction of economic goods and services in tandem with intelligent machines¹²⁶. Ultimately, understanding how human systems can be altered by the introduction of intelligent machines into our lives is a vital component of the study of machine behaviour.

Humans shape machine behaviour

Intelligent machines can alter human behaviour, and humans also create, inform and mould the behaviours of intelligent machines. We shape machine behaviours through the direct engineering of AI systems and through the training of these systems on both active human input and passive observations of human behaviours through the data that we create daily. The choice of which algorithms to use, what feedback to provide to those algorithms^{3,127} and on which data to train them are also, at present, human decisions and can directly alter machine behaviours. An important component in the study of machine behaviour is to understand how these engineering processes alter the resulting behaviours of AI, whether the training data are responsible for a particular behaviour of the machine, whether it is the algorithm itself or whether it is a combination of both algorithm and data. The framework outlined in Fig. 3 suggests that there will be complementary answers to the each of these questions. Examining how altering the parameters of the engineering process can alter the subsequent behaviours of intelligent machines as they interact with other machines and with humans in natural settings is central to a holistic understanding of machine behaviour.

Human–machine co-behaviour

Although it can be methodologically convenient to separate studies into the ways that humans shape machines and vice versa, most AI systems function in domains where they co-exist with humans in complex hybrid systems^{39,67,125,128}. Questions of importance to the study of these systems include those that examine the behaviours that characterize human–machine interactions including cooperation, competition and coordination—for example, how human biases combine with AI to alter human emotions or beliefs^{14,55,56,129,130}, how human tendencies

couple with algorithms to facilitate the spread of information⁵⁵, how traffic patterns can be altered in streets populated by large numbers of both driverless and human-driven cars and how trading patterns can be altered by interactions between humans and algorithmic trading agents²⁹ as well as which factors can facilitate trust and cooperation between humans and machines^{88,131}.

Another topic in this area relates to robotic and software-driven automation of human labour¹³². Here we see two different types of machine–human interactions. One is that machines can enhance a human's efficiency, such as in robotic- and computer-aided surgery. Another is that machines can replace humans, such as in driverless transportation and package delivery. This leads to questions about whether machines end up doing more of the replacing or the enhancing in the longer run and what human–machine co-behaviours will evolve as a result.

The above examples highlight that many of the questions that relate to hybrid human–machine behaviours must necessarily examine the feedback loops between human influence on machine behaviour and machine influence on human behaviour simultaneously. Scholars have begun to examine human–machine interactions in formal laboratory environments, observing that interactions with simple bots can increase human coordination³⁹ and that bots can cooperate directly with humans at levels that rival human–human cooperation¹³³. However, there remains an urgent need to further understand feedback loops in natural settings, in which humans are increasingly using algorithms to make decisions¹³⁴ and subsequently informing the training of the same algorithms through those decisions. Furthermore, across all types of questions in the domain of machine behavioural ecology, there is a need for studies that examine longer-run dynamics of these hybrid systems⁵³ with particular emphasis on the ways that human social interactions^{135,136} may be modified by the introduction of intelligent machines¹³⁷.

Outlook

Furthering the study of machine behaviour is critical to maximizing the potential benefits of AI for society. The consequential choices that we make regarding the integration of AI agents into human lives must be made with some understanding of the eventual societal implications of these choices. To provide this understanding and anticipation, we need a new interdisciplinary field of scientific study: machine behaviour.

For this field to succeed, there are a number of relevant considerations. First, studying machine behaviour does not imply that AI algorithms necessarily have independent agency nor does it imply algorithms should bear moral responsibility for their actions. If a dog bites someone, the dog's owner is held responsible. Nonetheless, it is useful to study the behavioural patterns of animals to predict such aberrant behaviour. Machines operate within a larger socio-technical fabric, and their human stakeholders are ultimately responsible for any harm their deployment might cause.

Second, some commentators might suggest that treating AI systems as agents occludes the focus on the underlying data that such AI systems are trained on. Indeed, no behaviour is ever fully separable from the environmental data on which that agent is trained or developed; machine behaviour is no exception. However, it is just as critical to understand how machine behaviours vary with altered environmental inputs as it is to understand how biological agents' behaviours vary depending on the environments in which they exist. As such, scholars of machine behaviour should focus on characterizing agent behaviour across diverse environments, much as behavioural scientists desire to characterize political behaviours across differing demographic and institutional contexts.

Third, machines exhibit behaviours that are fundamentally different from animals and humans, so we must avoid excessive anthropomorphism and zoomorphism. Even if borrowing existing behavioural scientific methods can prove useful for the study of machines, machines may exhibit forms of intelligence and behaviour that are qualitatively different—even alien—from those seen in biological agents.

Furthermore, AI scientists can dissect and modify AI systems more easily and more thoroughly than is the case for many living systems. Although parallels exist, the study of AI systems will necessarily differ from the study of living systems.

Fourth, the study of machine behaviour will require cross-disciplinary efforts^{82,103} and will entail all of the challenges associated with such research^{138,139}. Addressing these challenges is vital¹⁴⁰. Universities and governmental funding agencies can play an important part in the design of large-scale, neutral and trusted cross-disciplinary studies¹⁴¹.

Fifth, the study of machine behaviour will often require experimental intervention to study human-machine interactions in real-world settings^{142,143}. These interventions could alter the overall behaviour of the system, possibly having adverse effects on normal users¹⁴⁴. Ethical considerations such as these need careful oversight and standardized frameworks.

Finally, studying intelligent algorithmic or robotic systems can result in legal and ethical problems for researchers studying machine behaviour. Reverse-engineering algorithms may require violating the terms of service of some platforms; for example, in setting up fake personas or masking true identities. The creators or maintainers of the systems of interest could embroil researchers in legal challenges if the research damages the reputation of their platforms. Moreover, it remains unclear whether violating terms of service may expose researchers to civil or criminal penalties (for example, through the Computer Fraud and Abuse Act in the United States), which may further discourage this type of research¹⁴⁵.

Understanding the behaviours and properties of AI agents—and the effects they might have on human systems—is critical. Society can benefit tremendously from the efficiencies and improved decision-making that can come from these agents. At the same time, these benefits may falter without minimizing the potential pitfalls of the incorporation of AI agents into everyday human life.

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